



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: REAL TIME OPERATING SYSTEMS – (24IN3103HF)

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TITLE OF THE PROJECT	Edge AI Predictive Maintenance Controller for Industry 5.0	
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Abstract:

The increasing adoption of smart and connected manufacturing systems in Industry 5.0 requires intelligent maintenance solutions capable of monitoring industrial equipment continuously and predicting potential failures before they result in unexpected downtime. Conventional preventive maintenance approaches generally depend on fixed maintenance schedules and manual inspection, which may not accurately represent the real-time condition of industrial machinery. Such approaches can lead to unnecessary maintenance, increased operational costs, and unexpected equipment failures. Therefore, there is a need for a real-time predictive maintenance controller that combines Edge Artificial Intelligence, real-time computing, and industrial communication technologies to achieve fast and intelligent machine-health monitoring.

To address this need, the proposed project develops an **Edge AI Predictive Maintenance Controller for Industry 5.0** using **RT-Linux, TensorFlow Lite, MQTT, Python, Docker, and Grafana**. The system continuously collects machine parameters such as vibration, temperature, current, and rotational speed through sensors connected to an edge computing

platform. **RT-Linux** is employed to provide real-time task execution and scheduling for critical operations such as sensor acquisition, signal processing, AI inference, and controller response. This enables the system to perform time-sensitive monitoring and control operations with reduced latency.

The acquired sensor data is processed locally using **Python-based signal-processing and machine-learning pipelines**. Relevant features are extracted from the sensor signals and provided to a lightweight **TensorFlow Lite Edge AI model** for machine-condition classification and predictive maintenance analysis. The AI model identifies operating conditions such as normal, warning, and potential-failure states without requiring continuous transmission of raw sensor data to a remote cloud server. This edge-based approach reduces communication latency and improves the responsiveness of the maintenance controller.

For industrial communication, **MQTT** is used as a lightweight publish-subscribe messaging protocol to transfer machine-health information, AI predictions, alarms, and operational parameters between the edge controller and monitoring services. The AI inference and supporting services are packaged using **Docker**, providing an isolated and reproducible deployment environment. **Grafana** is integrated to provide a real-time visualization dashboard containing machine parameters, health status, predicted faults, sensor trends, and maintenance alerts. This enables operators to monitor machine conditions and make timely maintenance decisions.

The proposed system integrates **Edge AI, RT-Linux real-time computing, TensorFlow Lite, MQTT communication, Python processing, Docker-based deployment, and Grafana visualization** into a unified predictive maintenance framework. The expected outcome is a low-latency and scalable intelligent maintenance controller capable of detecting abnormal machine conditions at an early stage, reducing unplanned downtime, improving equipment reliability, and supporting human-centric and intelligent manufacturing in Industry 5.0 environments.

Project Architecture/Project Overview:

